Costa's Minimal Surface

Xiaoshuo Lin

University of Science and Technology of China

August 23, 2024

Outline

Producing Minimal Surfaces

The Two Weierstrass Functions

Parametrization of Costa's Minimal Surface

Asymptotics of the Three Ends

The Weierstrass Representation

Theorem (The Weierstrass Representation Theorem)

Let f and g be functions on a simply connected domain $U \subset \mathbb{C}$, where g is meromorphic and f is holomorphic, such that wherever g has a pole of order m, f has a zero of order at least $2m$ (or equivalently, such that the product fg^2 is holomorphic). Fix $z_0 \in U$, and let c_1, c_2, c_3 be constants. Then the surface with coordinates (x_1, x_2, x_3) is ***minimal****, where the x_k are defined as follows:*

$$x_k(z) = \mathfrak{Re}\left\{\int_{z_0}^{z} \varphi_k(w)\,\mathrm{d}w\right\} + c_k, \quad k = 1, 2, 3.$$

$$\varphi_1 = \frac{f\left(1 - g^2\right)}{2}, \quad \varphi_2 = \frac{\mathrm{i}f\left(1 + g^2\right)}{2}, \quad \varphi_3 = fg.$$

A Basic Example

From the functions

$$f(z) = -e^{-z} \text{ and } g(z) = -e^{z}$$

we obtain (up to constants)

$$\begin{cases} x_1(u, v) = \cosh u \cos v, \\ x_2(u, v) = \cosh u \sin v, \\ x_3(u, v) = u, \end{cases}$$

which describes the catenoid.

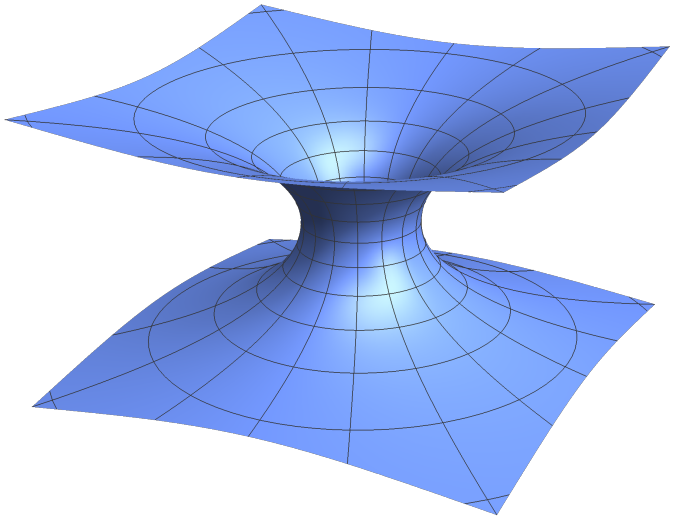


Figure: A catenoid

The Weierstrass $\wp$ and ζ

We choose the lattice $\mathbb{Z}[\mathrm{i}] = \{m + \mathrm{i}n : m, n \in \mathbb{Z}\}$ so that the Weierstrass $\wp$ and ζ functions are defined by

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right),$$

$$\zeta(z) = \frac{1}{z} + \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \left(\frac{1}{z-\omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right).$$

Clearly, $\zeta'(z) = -\wp(z)$. Let us denote

$$e_1 = \wp(\tfrac{1}{2}), \quad e_2 = \wp\left(\tfrac{\mathrm{i}}{2}\right), \quad e_3 = \wp\left(\tfrac{1+\mathrm{i}}{2}\right).$$

Please don't confuse ζ with the Riemann zeta function.

Identities of $\wp$ and ζ

1. $\wp(z+m+\mathrm{i}n) = \wp(z)$ for all $m, n \in \mathbb{Z}$.
2. $\wp(z_1+z_2) = \frac{1}{4}\left(\frac{\wp'(z_1)-\wp'(z_2)}{\wp(z_1)-\wp(z_2)}\right)^2 - \wp(z_1) - \wp(z_2)$.
3. $\wp'(z)^2 = \left[4\wp(z)^2 - g_2\right]\wp(z)$.
4. $\wp'(z)^2 = 4\wp(z)\left[\wp(z)^2 - e_1^2\right]$.
5. $\wp\left(z+\frac{1}{2}\right) = e_1 + \frac{2e_1^2}{\wp(z)-e_1}$.
6. $\wp\left(z+\frac{\mathrm{i}}{2}\right) = e_2 + \frac{2e_2^2}{\wp(z)-e_2} = -e_1 + \frac{2e_1^2}{\wp(z)+e_1}$.
7. $\wp\left(z-\frac{1}{2}\right) - \wp\left(z-\frac{\mathrm{i}}{2}\right) - 2e_1 = \frac{16e_1^3\wp(z)}{\wp'(z)^2}$.

Identities of $\wp$ and ζ

8. $\zeta(z+m+\mathrm{i}n) = \zeta(z) + 2m\zeta\left(\frac{1}{2}\right) + 2n\zeta\left(\frac{\mathrm{i}}{2}\right)$ for all $m, n \in \mathbb{Z}$.
9. $\mathrm{i}\zeta(\mathrm{i}z) = \zeta(z)$.
10. $\zeta\left(\frac{1}{2}\right) = \mathrm{i}\zeta\left(\frac{\mathrm{i}}{2}\right) = \frac{\pi}{2}$.
11. $\zeta\left(\frac{1+\mathrm{i}}{2}\right) = \frac{(1-\mathrm{i})\pi}{2}$.
12. $\zeta(z+u) - \zeta(z) - \zeta(u) = \frac{1}{2}\frac{\wp'(z)-\wp'(u)}{\wp(z)-\wp(u)}$.

1. Clear from the definition of $\wp$.
2. A well-known addition formula that can be found in most textbooks on elliptic functions. So is 12.
3. Corollary 2.3 in Chapter 9 of *Complex Analysis* by Stein and Shakarchi gives the identity

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where

$$g_2 = 60 \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \frac{1}{\omega^4} \quad \text{and} \quad g_3 = 140 \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \frac{1}{\omega^6}.$$

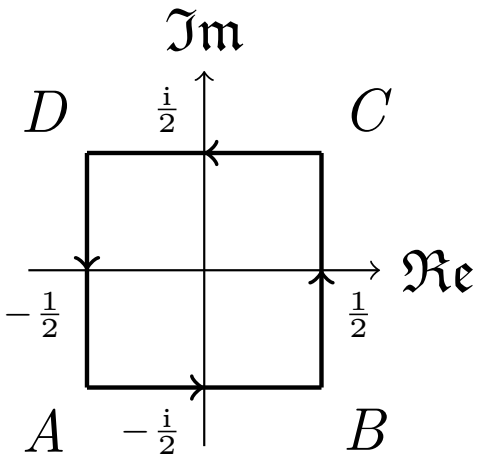
In our case, $g_3 = 0$ since

$$(m - \mathrm{i}n)^6 + (m + \mathrm{i}n)^6 + (n - \mathrm{i}m)^6 + (n + \mathrm{i}m)^6 = 0.$$

4. It is known that $1/2$, $\mathrm{i}/2$ and $(1+\mathrm{i})/2$ are the roots of the cubic polynomial $\left[4\wp(z)^2 - g_2\right]\wp(z)$, and $\wp\left(\frac{1+\mathrm{i}}{2}\right) = e_3 = 0$. Hence $4e_1^2 = g_2$, and the identity follows from 3.
5. Apply 2 and then 4.
6. Apply 2 and then 4. Note that $e_2 = -e_1$.
7. By 1 $\wp\left(z - \frac{1}{2}\right) - \wp\left(z - \frac{\mathrm{i}}{2}\right) = \wp\left(z + \frac{1}{2}\right) - \wp\left(z + \frac{\mathrm{i}}{2}\right)$. Then combine 5 and 6 to get the identity.
8. Since $\zeta'(z) = -\wp(z)$ and $\wp(z+1) = \wp(z)$, the two functions $\zeta(z+1)$ and $\zeta(z)$ differ by a constant, say $\zeta(z+1) = \zeta(z) + c$. Take $z = -\frac{1}{2}$ and use the fact that ζ is odd to get $c = 2\zeta\left(\frac{1}{2}\right)$. The same argument gives $\zeta(z+\mathrm{i}) = \zeta(z) + 2\zeta\left(\frac{\mathrm{i}}{2}\right)$.
9. Clear from the definition of ζ and the fact that $\mathrm{i}\mathbb{Z}[\mathrm{i}] = \mathbb{Z}[\mathrm{i}]$.

10. The residue theorem gives

$$\int_{ABCDA} \zeta(z)\,\mathrm{d}z = 2\pi\mathrm{i}.$$



On the other hand, by 8 we have

$$\int_{CD} \zeta(z)\,\mathrm{d}z = \int_{BA} \zeta(z)\,\mathrm{d}z - 2\zeta\left(\tfrac{\mathrm{i}}{2}\right), \quad \int_{BC} \zeta(z)\,\mathrm{d}z = \int_{AD} \zeta(z)\,\mathrm{d}z + 2\mathrm{i}\zeta\left(\tfrac{1}{2}\right).$$

Combining these equations gives $\zeta\left(\frac{1}{2}\right) + \mathrm{i}\zeta\left(\frac{\mathrm{i}}{2}\right) = \pi$. Then use 9.

11. Take $z = -\frac{1+\mathrm{i}}{2}$ and $m = n = 1$ in 8 and use the fact that ζ is odd to get $\zeta\left(\frac{1+\mathrm{i}}{2}\right) = \zeta\left(\frac{1}{2}\right) + \zeta\left(\frac{\mathrm{i}}{2}\right)$. Then 10 applies.

The Weierstrass Data

Costa's minimal surface is defined as a Weierstrass patch using the functions

$$f(z) = \wp(z) \quad \text{and} \quad g(z) = \frac{A}{\wp'(z)}.$$

In order that Costa's minimal surface has no self-intersections, we need to take[a]

$$A = 2\sqrt{2\pi}\, e_1 \approx 34.46707.$$

[a] D. Hoffman and W. Meeks, A complete minimal surface in $\mathbb{R}^3$ with genus one and three ends, *J. Differential Geometry 21* (1985), 109–127.

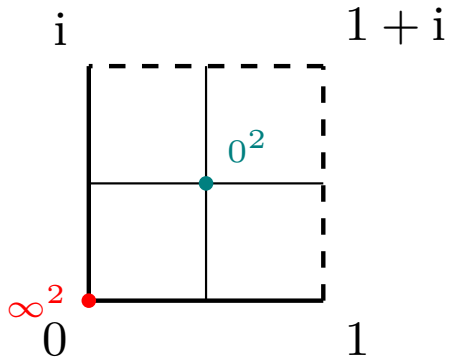


Figure: Zeros and poles of f

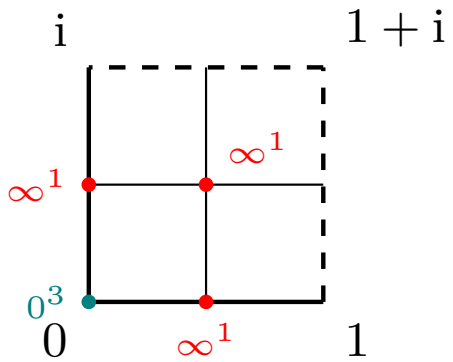


Figure: Zeros and poles of g

We shall use ζ to express the coordinates without integrals.

Using 7 we obtain

$$
\begin{aligned}
f(w)\left[1-g(w)^2\right] &= \wp(w) - \frac{A^2\wp(w)}{\wp'(w)^2} \\
&= \wp(w) - \frac{A^2}{16e_1^3}\left[\wp\left(w-\tfrac{1}{2}\right) - \wp\left(w-\tfrac{\mathrm{i}}{2}\right) - 2e_1\right] \\
&= \wp(w) - \frac{\pi}{2e_1}\left[\wp\left(w-\tfrac{1}{2}\right) - \wp\left(w-\tfrac{\mathrm{i}}{2}\right) - 2e_1\right] \\
&= \wp(w) + \pi - \frac{\pi}{2e_1}\left[\wp\left(w-\tfrac{1}{2}\right) - \wp\left(w-\tfrac{\mathrm{i}}{2}\right)\right].
\end{aligned}
$$

Take $z_0 = \frac{1+\mathrm{i}}{2}$. Integrating both sides and using 10 and 11, we get

$$\begin{aligned}
&\int_{z_0}^{z} f(w)\left[1-g(w)^2\right]\mathrm{d}w \\
&= \left\{-\zeta(w)+\pi w+\frac{\pi}{2e_1}\left[\zeta\left(w-\tfrac{1}{2}\right)-\zeta\left(w-\tfrac{\mathrm{i}}{2}\right)\right]\right\}\Bigg|_{z_0}^{z} \\
&= -\zeta(z)+\pi z+\frac{\pi}{2e_1}\left[\zeta\left(z-\tfrac{1}{2}\right)-\zeta\left(z-\tfrac{\mathrm{i}}{2}\right)\right] \\
&\quad +\zeta\left(\tfrac{1+\mathrm{i}}{2}\right)-\frac{\pi(1+\mathrm{i})}{2}-\frac{\pi}{2e_1}\left[\zeta\left(\tfrac{\mathrm{i}}{2}\right)-\zeta\left(\tfrac{1}{2}\right)\right] \\
&= -\zeta(z)+\pi z+\frac{\pi}{2e_1}\left[\zeta\left(z-\tfrac{1}{2}\right)-\zeta\left(z-\tfrac{\mathrm{i}}{2}\right)\right]-\mathrm{i}\pi+\frac{\pi^2(1+\mathrm{i})}{4e_1}.
\end{aligned}$$

Dividing by 2 and taking the real part, we get x_1.

Similarly,

$$f(w)\left[1+g(w)^2\right]=\wp(w)-\pi+\frac{\pi}{2e_1}\left[\wp\left(w-\tfrac{1}{2}\right)-\wp\left(w-\tfrac{\mathrm{i}}{2}\right)\right]$$

and then

$$\begin{aligned}&\int_{z_0}^{z} f(w)\left[1+g(w)^2\right]\mathrm{d}w\\ &=\left\{-\zeta(w)-\pi w-\frac{\pi}{2e_1}\left[\zeta\left(w-\tfrac{1}{2}\right)-\zeta\left(w-\tfrac{\mathrm{i}}{2}\right)\right]\right\}\Bigg|_{z_0}^{z}\\ &=-\zeta(z)-\pi z-\frac{\pi}{2e_1}\left[\zeta\left(z-\tfrac{1}{2}\right)-\zeta\left(z-\tfrac{\mathrm{i}}{2}\right)\right]+\pi-\frac{\pi^2(1+\mathrm{i})}{4e_1}.\end{aligned}$$

From this we can find x_2.

Using 4 we obtain

$$\begin{aligned}
\int_{z_0}^{z} f(w)g(w)\,\mathrm{d}w &= A\int_{z_0}^{z}\frac{\wp(w)}{\wp'(w)}\,\mathrm{d}w = \frac{A}{4}\int_{z_0}^{z}\frac{\wp'(w)\,\mathrm{d}w}{\wp(w)^2-e_1^2}\\
&= \frac{A}{8e_1}\int_{z_0}^{z}\left(\frac{\wp'(w)}{\wp(w)-e_1}-\frac{\wp'(w)}{\wp(w)+e_1}\right)\mathrm{d}w\\
&= \frac{\sqrt{2\pi}}{4}\log\left(\frac{\wp(w)-e_1}{\wp(w)+e_1}\right)\Bigg|_{z_0}^{z}\\
&= \frac{\sqrt{2\pi}}{4}\left\{\log\left(\frac{\wp(z)-e_1}{\wp(z)+e_1}\right)-\log\left(\frac{e_3-e_1}{e_3+e_1}\right)\right\}\\
&= \frac{\sqrt{2\pi}}{4}\left\{\log\left(\frac{\wp(z)-e_1}{\wp(z)+e_1}\right)-\pi\mathrm{i}\right\}.
\end{aligned}$$

Taking the real part gives x_3.

Parametric Equations

Costa's minimal surface is given by (x_1, x_2, x_3) where

$$\begin{cases} x_1(u,v) = \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u+\mathrm{i}v) + \pi u + \frac{\pi^2}{4e_1} \right. \\ \qquad\qquad \left. + \frac{\pi}{2e_1}\left[\zeta\left(u+\mathrm{i}v-\frac{1}{2}\right) - \zeta\left(u+\mathrm{i}v-\frac{\mathrm{i}}{2}\right)\right]\right\}, \\ x_2(u,v) = \frac{1}{2}\mathfrak{Re}\left\{-\mathrm{i}\zeta(u+\mathrm{i}v) + \pi v + \frac{\pi^2}{4e_1} \right. \\ \qquad\qquad \left. - \frac{\pi\mathrm{i}}{2e_1}\left[\zeta\left(u+\mathrm{i}v-\frac{1}{2}\right) - \zeta\left(u+\mathrm{i}v-\frac{\mathrm{i}}{2}\right)\right]\right\}, \\ x_3(u,v) = \frac{\sqrt{2\pi}}{4}\log\left|\frac{\wp(u+\mathrm{i}v) - e_1}{\wp(u+\mathrm{i}v) + e_1}\right|. \end{cases}$$

What the Surface Looks Like

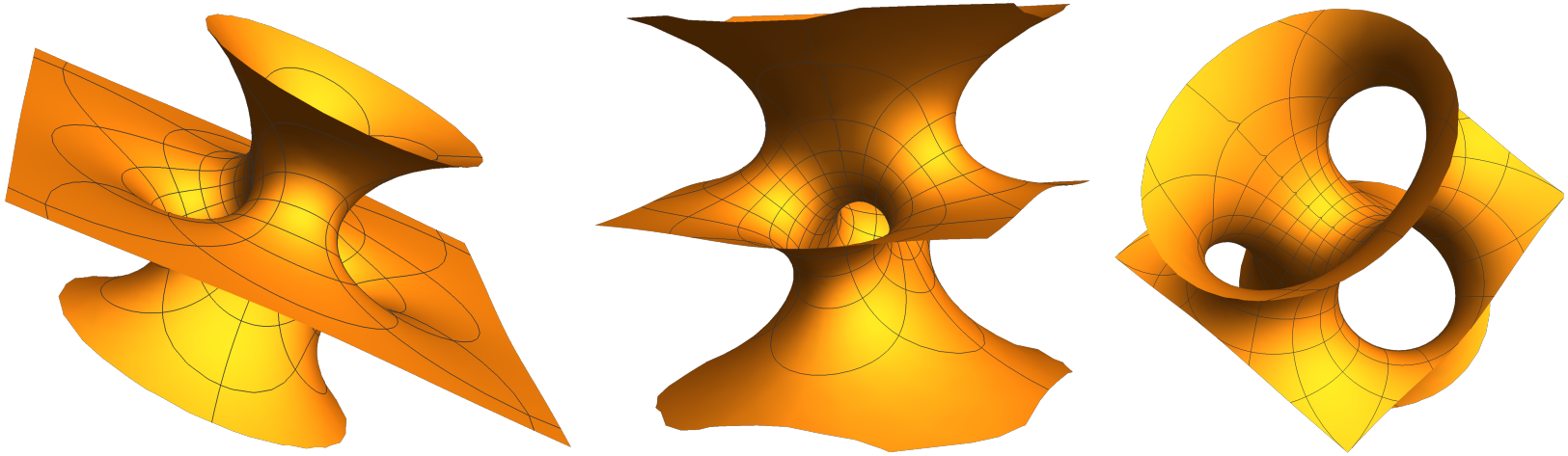


Figure: Zoom in on the Costa's surface (left to right)

Domain of the Parameters

By 8 and 10 we have $\zeta(z+1) = \zeta(z) + \pi$, hence

$$\begin{aligned} x_1(u+1, v) =& \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u+\mathrm{i}v) - \pi + \pi u + \pi + \frac{\pi^2}{4e_1}\right. \\ &\left. + \frac{\pi}{2e_1}\left[\zeta\left(u+\mathrm{i}v-\tfrac{1}{2}\right) + \pi - \zeta\left(u+\mathrm{i}v-\tfrac{\mathrm{i}}{2}\right) - \pi\right]\right\} \\ =& x_1(u, v). \end{aligned}$$

Similarly, one can show that $x_1(u, v+1) = x_1(u, v)$ and

$$\begin{aligned} x_2(u+1, v) &= x_2(u, v), \quad & x_2(u, v+1) &= x_2(u, v), \\ x_3(u+1, v) &= x_3(u, v), \quad & x_3(u, v+1) &= x_3(u, v). \end{aligned}$$

Therefore, we may restrict u and v to the unit square $[0, 1) \times [0, 1)$.

Two Catenoidal Ends and One Planar End

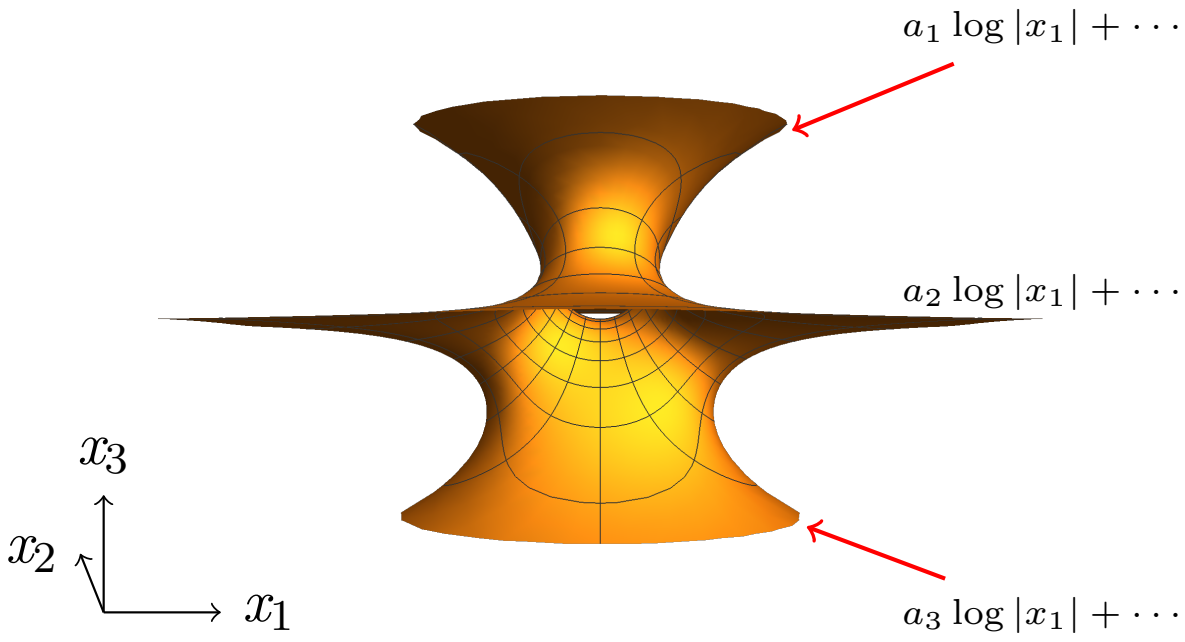


Figure: Front view of the Costa's surface

Key Observations (i)

- $\wp(x) \in \mathbb{R}$ whenever $x \in \mathbb{R}$.

$$\overline{\wp(x)} = \frac{1}{x^2} + \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \left(\frac{1}{(x-\overline{\omega})^2} - \frac{1}{\overline{\omega}^2} \right) = \wp(x).$$

- $\zeta(x) \in \mathbb{R}$ whenever $x \in \mathbb{R}$.

$$\overline{\zeta(x)} = \frac{1}{x} + \sum_{\omega \in \mathbb{Z}[\mathrm{i}]\setminus\{0\}} \left(\frac{1}{x-\overline{\omega}} + \frac{1}{\overline{\omega}} + \frac{x}{\overline{\omega}^2} \right) = \zeta(x).$$

Key Observations (ii)

$$x_2(u,0) = \frac{1}{2}\mathfrak{Re}\left\{-\mathrm{i}\zeta(u) + \frac{\pi^2}{4e_1} - \frac{\pi\mathrm{i}}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right) - \zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right]\right\}$$
$$= \frac{1}{2}\left\{\frac{\pi^2}{4e_1} - \frac{\pi}{2e_1}\mathfrak{Im}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\}\right\}.$$

By 12 we have

$$\zeta\left(u-\tfrac{\mathrm{i}}{2}\right) = \zeta(u) + \zeta\left(-\tfrac{\mathrm{i}}{2}\right) + \frac{1}{2}\frac{\wp'(u) - \wp'\left(\frac{\mathrm{i}}{2}\right)}{\wp(u) - \wp\left(\frac{\mathrm{i}}{2}\right)}.$$

Since $\wp\left(\frac{\mathrm{i}}{2}\right) = e_2 = -e_1 \in \mathbb{R}$ and $\wp'\left(\frac{\mathrm{i}}{2}\right) = 0$, we obtain

$$\mathfrak{Im}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\} = \mathfrak{Im}\left\{\zeta\left(-\tfrac{\mathrm{i}}{2}\right)\right\} = \mathfrak{Im}\left\{\tfrac{\pi\mathrm{i}}{2}\right\} = \tfrac{\pi}{2}.$$

Hence $x_2(u,0) = 0$.

Key Observations (iii)

$$
\begin{aligned}
& x_2\left(u, \tfrac{1}{2}\right) \\
=& \frac{1}{2}\mathfrak{Re}\left\{-\mathrm{i}\zeta\left(u+\tfrac{\mathrm{i}}{2}\right)+\frac{\pi}{2}+\frac{\pi^2}{4e_1}-\frac{\pi\mathrm{i}}{2e_1}\left[\zeta\left(u+\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)-\zeta(u)\right]\right\} \\
=& \frac{1}{2}\left\{\mathfrak{Im}\left\{\zeta\left(u+\tfrac{\mathrm{i}}{2}\right)\right\}+\frac{\pi}{2}+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\mathfrak{Im}\left\{\zeta\left(u+\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)\right\}\right\}.
\end{aligned}
$$

As in the case with (ii), we have

$$
\begin{aligned}
\mathfrak{Im}\left\{\zeta\left(u+\tfrac{\mathrm{i}}{2}\right)\right\} &= \mathfrak{Im}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)-\pi\mathrm{i}\right\} = \frac{\pi}{2}-\pi = -\frac{\pi}{2}, \\
\mathfrak{Im}\left\{\zeta\left(u+\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)\right\} &= \mathfrak{Im}\left\{\zeta\left(\left(u-\tfrac{1}{2}\right)+\tfrac{\mathrm{i}}{2}\right)\right\} = -\frac{\pi}{2}.
\end{aligned}
$$

Therefore, $x_2\left(u, \frac{1}{2}\right) = 0$.

Key Observations (iv)

- $x_1(u,0) \to -\infty$ as $u \searrow 0$.
- $x_1(u,0) \to -\infty$ as $u \nearrow \frac{1}{2}$.
- $x_1(u,0) \to +\infty$ as $u \searrow \frac{1}{2}$.
- $x_1(u,0) \to +\infty$ as $u \nearrow 1$.
- $x_3(u,0) \to 0$ as $u \to 0$.
- $x_3(u,0) \to -\infty$ as $u \to \frac{1}{2}$.

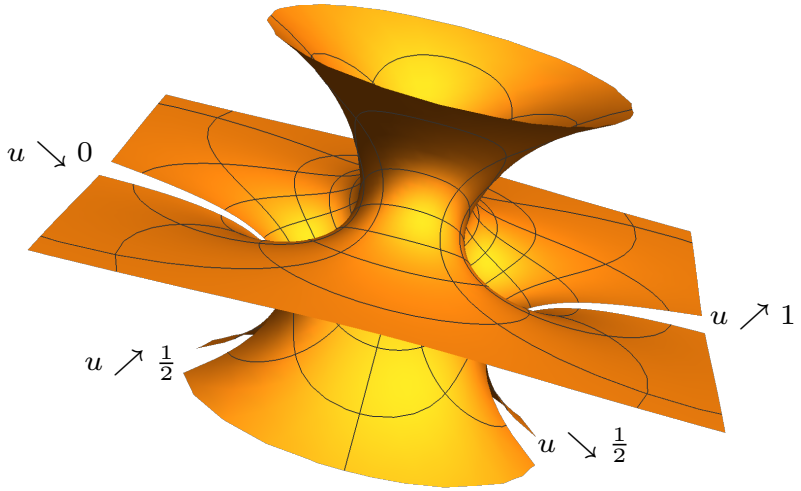


Figure: The curve $v = 0$

Proof of (iv)

When $u \searrow 0$,

$$
\begin{aligned}
& x_1(u,0) \\
=& \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right]\right\} \\
=& \frac{1}{2}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{1}{2}\left\{-\frac{1}{u}+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& -\frac{1}{2u} \to -\infty.
\end{aligned}
$$

Proof of (iv)

When $u \nearrow \frac{1}{2}$,

$$
\begin{aligned}
& x_1(u,0) \\
=& \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right]\right\} \\
=& \frac{1}{2}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{1}{2}\left\{-\zeta\left(\tfrac{1}{2}\right)+\frac{\pi}{2}+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(\tfrac{1-\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{\pi}{4e_1}\zeta\left(u-\tfrac{1}{2}\right) \\
\sim& \frac{\pi}{4e_1}\frac{1}{u-\frac{1}{2}} \to -\infty.
\end{aligned}
$$

Proof of (iv)

When $u \searrow \frac{1}{2}$,

$$
\begin{aligned}
& x_1(u,0) \\
=& \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right]\right\} \\
=& \frac{1}{2}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{1}{2}\left\{-\zeta\left(\tfrac{1}{2}\right)+\frac{\pi}{2}+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(\tfrac{1-\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{\pi}{4e_1}\zeta\left(u-\tfrac{1}{2}\right) \\
\sim& \frac{\pi}{4e_1}\frac{1}{u-\frac{1}{2}} \to +\infty.
\end{aligned}
$$

Proof of (iv)

When $u \nearrow 1$,

$$
\begin{aligned}
& x_1(u,0) \\
=& \frac{1}{2}\mathfrak{Re}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right]\right\} \\
=& \frac{1}{2}\left\{-\zeta(u)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u-\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(u-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& \frac{1}{2}\left\{-\frac{1}{u-1}-\pi+\pi+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(\tfrac{1}{2}\right)-\mathfrak{Re}\left\{\zeta\left(1-\tfrac{\mathrm{i}}{2}\right)\right\}\right]\right\} \\
\sim& -\frac{1}{2(u-1)} \to +\infty.
\end{aligned}
$$

Proof of (iv)

$$x_3(u,0) = \frac{\sqrt{2\pi}}{4} \log \left| \frac{\wp(u) - e_1}{\wp(u) + e_1} \right| .$$

- Since $\wp(u) \sim \frac{1}{u^2}$ as $u \to 0$, we have $x_3(u,0) \to 0$ as $u \to 0$.
- Since $\wp\left(\frac{1}{2}\right) = e_1$, we have $x_3(u,0) \to -\infty$ as $u \to \frac{1}{2}$.

The Coefficients a_2 and a_3

It is obvious from observation (iv) that $a_2 = 0$.
To find a_3, first note that

$$a_3 = \lim_{u \searrow \frac{1}{2}} \frac{x_3(u,0)}{\log x_1(u,0)} = \frac{\sqrt{2\pi}}{4} \lim_{u \searrow 0} \frac{\log \left| \frac{\wp(u+1/2)-e_1}{\wp(u+1/2)+e_1} \right|}{\log \frac{\pi}{4e_1 u}}.$$

Using 5 we have

$$\frac{\wp\left(u+\frac{1}{2}\right)-e_1}{\wp\left(u+\frac{1}{2}\right)+e_1} = 1 - \frac{2e_1}{\wp\left(u+\frac{1}{2}\right)+e_1} = 1 - \frac{2e_1}{2e_1 + \frac{2e_1^2}{\wp(u)-e_1}}$$
$$= \frac{e_1}{\wp(u)} \sim e_1 u^2 \quad \text{as } u \to 0.$$

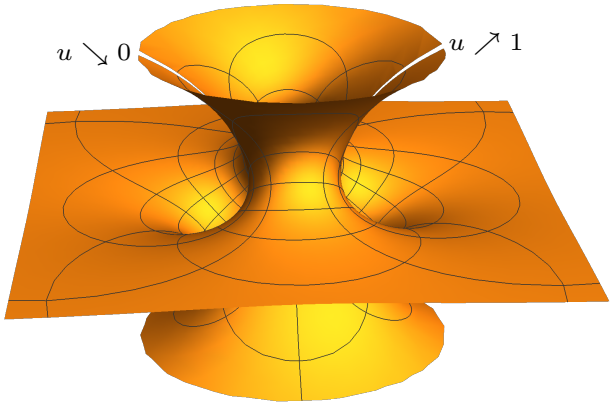
The Coefficients a_2 and a_3

Now

$$\begin{aligned} a_3 &= \frac{\sqrt{2\pi}}{4} \lim_{u \searrow 0} \frac{\log\left(e_1 u^2\right)}{\log \frac{\pi}{4 e_1 u}} \\ &= \frac{\sqrt{2\pi}}{4} \lim_{u \searrow 0} \frac{2 \log u + \log e_1}{-\log u + \log \frac{\pi}{4 e_1}} \\ &= -\sqrt{\frac{\pi}{2}} \approx -1.25331. \end{aligned}$$

Key Observations (v)

- $x_1\left(u, \frac{1}{2}\right) \to -\infty$ as $u \searrow 0$.
- $x_1\left(u, \frac{1}{2}\right) \to +\infty$ as $u \nearrow 1$.



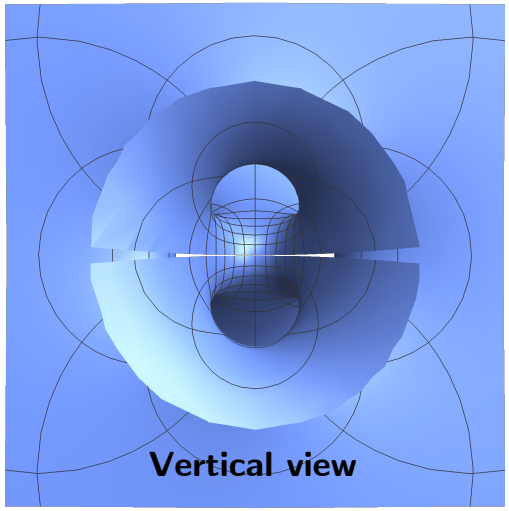


Figure: The curve $v = \frac{1}{2}$

Proof of (v)

$$
\begin{aligned}
&x_1\left(u, \tfrac{1}{2}\right) \\
=&\frac{1}{2}\mathfrak{Re}\left\{-\zeta\left(u+\tfrac{\mathrm{i}}{2}\right)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u+\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)-\zeta(u)\right]\right\}.
\end{aligned}
$$

When $u \searrow 0$,

$$
\begin{aligned}
x_1\left(u, \tfrac{1}{2}\right) &\sim \frac{1}{2}\mathfrak{Re}\left\{-\zeta\left(\tfrac{\mathrm{i}}{2}\right)+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)-\zeta(u)\right]\right\} \\
&\sim \frac{1}{2}\left\{\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left(-\frac{\pi}{2}-\frac{1}{u}\right)\right\} \\
&\sim -\frac{\pi}{4e_1 u} \to -\infty.
\end{aligned}
$$

Proof of (v)

$$
\begin{aligned}
&x_1\left(u, \tfrac{1}{2}\right) \\
=&\frac{1}{2}\mathfrak{Re}\left\{-\zeta\left(u+\tfrac{\mathrm{i}}{2}\right)+\pi u+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(u+\tfrac{\mathrm{i}}{2}-\tfrac{1}{2}\right)-\zeta(u)\right]\right\}.
\end{aligned}
$$

When $u \nearrow 1$,

$$
\begin{aligned}
x_1\left(u, \tfrac{1}{2}\right) &\sim \frac{1}{2}\mathfrak{Re}\left\{-\zeta\left(1+\tfrac{\mathrm{i}}{2}\right)+\pi+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\zeta\left(\tfrac{1+\mathrm{i}}{2}\right)-\zeta(u)\right]\right\} \\
&\sim \frac{1}{2}\left\{-\pi+\pi+\frac{\pi^2}{4e_1}+\frac{\pi}{2e_1}\left[\frac{\pi}{2}-\zeta(u-1)-\pi\right]\right\} \\
&\sim -\frac{\pi}{4e_1}\frac{1}{u-1} \to +\infty.
\end{aligned}
$$

The Coefficient a_1

As before, we write a_1 as

$$a_1 = \lim_{u \nearrow 1} \frac{x_3\left(u, \frac{1}{2}\right)}{\log x_1\left(u, \frac{1}{2}\right)} = \frac{\sqrt{2\pi}}{4} \lim_{u \nearrow 1} \frac{\log \left| \frac{\wp(u+\mathrm{i}/2)-e_1}{\wp(u+\mathrm{i}/2)+e_1} \right|}{\log \frac{\pi}{4e_1(1-u)}}.$$

Using 6 we have

$$\frac{\wp\left(u+\frac{\mathrm{i}}{2}\right)-e_1}{\wp\left(u+\frac{\mathrm{i}}{2}\right)+e_1} = 1 - \frac{2e_1}{\wp\left(u+\frac{\mathrm{i}}{2}\right)+e_1} = 1 - \frac{2e_1}{\frac{2e_1^2}{\wp(u)+e_1}}$$

$$= -\frac{\wp(u)}{e_1} \sim -\frac{1}{e_1(1-u)^2} \quad \text{as } u \to 1.$$

The Coefficient a_1

Now

$$\begin{aligned} a_1 &= \frac{\sqrt{2\pi}}{4} \lim_{u \nearrow 1} \frac{\log \frac{1}{e_1(1-u)^2}}{\log \frac{\pi}{4e_1(1-u)}} \\ &= \frac{\sqrt{2\pi}}{4} \lim_{u \nearrow 1} \frac{-2\log(1-u) - \log e_1}{-\log(1-u) + \log \frac{\pi}{4e_1}} \\ &= \sqrt{\frac{\pi}{2}} \approx 1.25331. \end{aligned}$$

The End